

How do frozen section results impact surgical decision-making and patient management in thyroid surgery, specifically regarding extent of thyroidectomy and lymph node dissection?

Frozen section analysis meaningfully impacts surgical decision-making in thyroid surgery when used selectively—demonstrating high utility for guiding extent of lymph node dissection in papillary thyroid carcinoma and for suspicious cytology cases where it modifies surgical plans in 16-65% of patients—but provides minimal benefit for follicular neoplasms or when preoperative cytology is already diagnostic.

Abstract

This systematic review of 40 studies demonstrates that the impact of frozen section on surgical decision-making in thyroid surgery is highly context-dependent. For lymph node assessment in clinically node-negative papillary thyroid carcinoma, frozen section consistently demonstrates high utility with sensitivities of 68.8-94.6% and specificities approaching 100% , enabling surgical plan modifications in 16.6-65% of cases . This approach allows appropriate extension from ipsilateral to bilateral central neck dissection while potentially reducing complications compared to routine bilateral dissection . For cytology suspicious for malignancy, frozen section leads to optimal operative procedures in 96% of cases with cost savings of \$474-1,298 per patient . Conversely, frozen section demonstrates limited utility for follicular neoplasms, with sensitivity ranging from 3.4-67% and 50-64% of cases deferred to permanent histology , resulting in surgical modification rates of only 0-3.4% and costs of \$7,800-12,470 per informative result .

The evidence supports selective rather than routine use of frozen section in thyroid surgery. Frozen section is most valuable when assessing nodal status in papillary thyroid carcinoma and when preoperative cytology is suspicious but not diagnostic for malignancy. It provides minimal benefit for follicular lesions, multinodular goiter , or when fine-needle aspiration is already diagnostic for papillary carcinoma . A consistent limitation across lymph node studies is the failure to detect micrometastases (≤ 2 mm) , though available follow-up data suggest these missed micrometastases have limited clinical significance with no recurrences reported .

Paper search

We performed a semantic search using the query "How do frozen section results impact surgical decision-making and patient management in thyroid surgery, specifically regarding extent of thyroidectomy and lymph node dissection?" across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 500 papers most relevant to the query.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Adult Population:** Does this study include adult patients (≥ 18 years) undergoing thyroid surgery?
- **Intraoperative Frozen Section:** Does this study involve intraoperative frozen section analysis of thyroid tissue?
- **Surgical Decision Outcomes:** Does this study report on surgical decision-making, extent of thyroidectomy, or lymph node dissection decisions?

- **Appropriate Study Design:** Is this study a randomized controlled trial, cohort study, case-control study, cross-sectional study, case series with ≥ 10 cases, systematic review, or meta-analysis?
- **Includes Intraoperative Analysis:** Does this study include intraoperative frozen section analysis (not solely final histopathology without frozen section)?
- **Surgical Decision Flexibility:** Does this study include patients where surgical extent was not predetermined (i.e., excludes studies with only known advanced/metastatic thyroid cancer patients where surgical decisions were already determined)?
- **Intraoperative Timing:** Was the frozen section analysis performed intraoperatively for surgical decision-making (not post-operatively or for research purposes only)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Study Population:**

Extract patient characteristics and inclusion criteria including:

- Number of patients
- Age (mean/median and range)
- Gender distribution
- Preoperative cytology results or molecular testing
- Clinical staging (cN0, cN1, etc.)
- Thyroid pathology types included
- Any exclusion criteria that might affect generalizability

- **Frozen Section Protocol:**

Extract details about how frozen section was performed including:

- When frozen section was performed (routine vs. selective)
- What tissue was examined (thyroid nodule, lymph nodes, or both)
- Criteria used for frozen section interpretation
- Who performed the frozen section (pathologist experience level)
- Time required for frozen section analysis
- Number of sections examined per case

- **Decision Algorithm:**

Extract the surgical decision-making framework including:

- Planned surgical approach before frozen section
- Specific criteria for changing surgical plan based on frozen section results
- Decision thresholds (e.g., size cutoffs, multifocality criteria)
- Who made intraoperative decisions (surgeon, multidisciplinary team)
- Alternative approaches compared (e.g., completion thyroidectomy vs. expedited final pathology)

- **Frozen Section Accuracy:**

Extract diagnostic performance metrics including:

- Sensitivity, specificity, positive and negative predictive values
- Accuracy compared to final pathology
- Number of cases where frozen section was deferred or inadequate
- Types of discordant cases (false positives, false negatives)
- Specific challenges noted (e.g., follicular lesions, micrometastases)
- Comparison to preoperative diagnostic methods if provided

• **Surgical Management Changes:**

Extract how frozen section results affected surgical decisions including:

- Number/percentage of cases where surgical plan was modified
- Types of modifications (lobectomy to total thyroidectomy, unilateral to bilateral lymph node dissection)
- Cases where frozen section led to appropriate vs. inappropriate changes
- Immediate vs. delayed completion surgeries required
- Cases where frozen section prevented unnecessary surgery

• **Clinical Outcomes:**

Extract patient outcomes and complications including:

- Final pathology results and cancer staging
- Rate of clinically significant cancers missed or detected
- Surgical complications (recurrent laryngeal nerve injury, hypoparathyroidism)
- Need for subsequent operations
- Length of hospital stay
- Patient quality of life measures if reported
- Long-term oncologic outcomes if available

• **Cost Analysis:**

Extract economic considerations including:

- Direct costs of frozen section procedure
- Cost comparisons between different management strategies
- Cost per case and total costs for study cohort
- Cost-effectiveness ratios if calculated
- Indirect costs considered (hospital stay, repeat surgery)
- Cost savings from avoiding unnecessary procedures

• **Key Findings:**

Extract the main conclusions and recommendations including:

- Authors' assessment of frozen section utility
- Specific recommendations for clinical practice
- Identified patient subgroups where frozen section may be most/least useful
- Comparison to alternative strategies
- Limitations acknowledged by authors
- Implications for surgical training or institutional protocols

Characteristics of Included Studies

This systematic review includes 40 studies examining the utility of frozen section analysis in thyroid surgery. The studies encompass a range of patient populations, thyroid pathologies, and clinical contexts.

Study	Full Text Retrieved?	Study Design	Sample Size	Population	Primary Pathology
Reema Mallick et al., 2019	No	Retrospective	236	83% female, mean age 55.6 years	Mixed thyroid nodules
M. Raffaelli et al., 2020	No	Prospective	94	cN0 unifocal PTC	Papillary thyroid carcinoma
M. Raffaelli et al., 2013	No	Prospective	48	cN0 unifocal PTC	Papillary thyroid carcinoma
Min Jhi Kim et al., 2021	No	Retrospective	290	Unilateral cN0 PTC	Papillary thyroid carcinoma
C. Bollig et al., 2018	No	Case series	65	Suspicious for malignancy cytology	Mixed
J. Roach et al., 2002	No	Retrospective	199	Indeterminate cytology	Mixed
Yelda Jozaghi et al., 2013	Yes	Prospective	300	Mean age 48.9 years, 256 women	Well-differentiated thyroid carcinoma
R. Králik et al., 2022	No	Retrospective	489	Median age 50 years	Differentiated thyroid carcinoma
Ryan W. Berg et al., 2016	No	Retrospective	192	Benign to suspicious cytology	Mixed
R. Udelsman et al., 2001	No	Randomized prospective	61	Follicular neoplasm on FNA	Follicular neoplasms
Y. Park et al., 2014	No	Prospective	268	PTC patients	Papillary thyroid carcinoma
Chaoyu Jiang et al., 2023	No	Retrospective	364	PTC patients	Papillary thyroid carcinoma
S. Abu-Ghanem et al., 2016	No	Case series	47	Bethesda V cytology	Mixed

Study	Full Text Retrieved?	Study Design	Sample Size	Population	Primary Pathology
S. Olson et al., 2006	No	Retrospective	236	Multinodular goiter	Multinodular thyroid disease
B. Chae et al., 2011	No	Prospective	203	Unilateral PTC, cN0	Papillary thyroid carcinoma
M. Raffaelli et al., 2015	No	Prospective	100	cN0 unifocal PTC	Papillary thyroid carcinoma
M. Raffaelli et al., 2020a	Yes	Prospective	60	cN0 PTC, mean age 40 years	Papillary thyroid carcinoma
M. Neale et al., 1993	No	Retrospective	735	Follicular neoplasms	Follicular neoplasms
Lei Zhang et al., 2007	No	Retrospective	1,057	Thyroid nodular disease	Mixed
M. Haymart et al., 2008	No	Retrospective	102	Diagnostic/suspicion for PTC	Papillary thyroid carcinoma
D. Hartl et al., 2014	No	Retrospective	295	cN0 PTC	Papillary thyroid carcinoma
A. Shaha et al., 1992	No	Retrospective	400	Age 18-88 years	Papillary/follicular carcinoma
R. Callcut et al., 2006	No	Retrospective	152	Mean age 47 years, 76% female	Follicular neoplasms
Li Peng et al., 2023	No	Retrospective	275	PTC with suspected LNM	Papillary thyroid carcinoma
A. Laurino et al., 2024	No	Retrospective	110	cN0 unifocal PTC	Papillary thyroid carcinoma
F. Lumachi et al., 2004	No	Retrospective	606	Median age 44 years, 80.5% women	Mixed
J. Paramo et al., 2008	No	Retrospective	71	Follicular neoplasm	Follicular neoplasms
David F. Schneider et al., 2017	No	Retrospective	639	Median age 52 years, 78.4% female	Indeterminate thyroid nodules
Yihao Liu et al., 2023	No	Retrospective multicenter	2,092	PTC patients	Papillary thyroid carcinoma

Study	Full Text Retrieved?	Study Design	Sample Size	Population	Primary Pathology
Dena Arumugam et al., 2016	Yes	Retrospective	148	Mean age 51 years, Bethesda IV	Follicular neoplasms
R. Králik et al., 2024	No	Retrospective	64	Median age 59 years, MTC	Medullary thyroid carcinoma
Luis Miguel Arciniegas et al., 2023	No	Prospective	154	cN0 PTC	Papillary thyroid carcinoma
L. Saydam et al., 2003	No	Retrospective	67	Mean age 44 years	Mixed
Christopher R. McHenry et al., 1996	No	Prospective	76	Adequate FNAB	Mixed
Christopher Wong et al., 2000	No	Case series	155	Solitary thyroid nodule	Mixed
A. Bastagli et al., 2001	No	Retrospective	155	Follicular neoplasms	Follicular neoplasms
C. Bollig et al., 2018a	No	Case series	48	Nodules >4 cm, nonmalignant cytology	Mixed
H. Moon et al., 2009	No	Retrospective	217	Suspicious for PTC on US-FNAB	Papillary thyroid carcinoma
M. J. Kim et al., 2018	No	Retrospective	265	PTC patients	Papillary thyroid carcinoma
O. Prajapati et al., 2015	Yes	Prospective	268	PTC patients	Papillary thyroid carcinoma

The included studies demonstrate considerable heterogeneity in study populations, with sample sizes ranging from 47 to 2,092 patients. The majority of studies focused on papillary thyroid carcinoma (22 studies), followed by follicular neoplasms (7 studies), and mixed pathologies (11 studies). Most studies were retrospective in design, with only a minority employing prospective or randomized methodology. Full text was available for only 4 studies, with the remainder analyzed through abstracts only.

Frozen Section Protocols and Tissue Examined

Studies varied considerably in their frozen section protocols, including which tissues were examined and the criteria used for interpretation.

Study	Tissue Examined	FS Timing	Interpretation Criteria
M. Raffaelli et al., 2020	Ipsilateral central neck nodes	Routine intraoperative	Presence of metastases
M. Raffaelli et al., 2013	Ipsilateral central compartment lymph nodes	Routine	Not specified
Min Jhi Kim et al., 2021	Central lymph nodes	Intraoperative	Cut-off size 5 mm
Yelda Jozaghi et al., 2013	Sentinel lymph nodes	Routine	Presence of metastasis
R. Králik et al., 2022	Lymph nodes of central compartment and thyroid gland	Not specified	Not specified
Y. Park et al., 2014	Thyroid tissue	Intraoperative	ETE evaluation
Chaoyu Jiang et al., 2023	Both thyroid and lymph nodes	Routine	>5 LNM or 1-5 LNM with ETE/multifocality
B. Chae et al., 2011	Ipsilateral central lymph node	Routine	Presence of malignancy
M. Raffaelli et al., 2015	Lymph nodes	Routine	Presence of metastases
M. Raffaelli et al., 2020a	Ipsilateral central neck nodes	Routine	Metastases >2 mm
D. Hartl et al., 2014	Lymph nodes levels III and IV	Selective	Not specified
Li Peng et al., 2023	Central lymph nodes	Intraoperative	Maximum diameter >2.05 mm, number >5.5, ratio >0.53
Dena Arumugam et al., 2016	Thyroid nodule	Routine	Nuclear features for PTC, capsular/vascular invasion for FTC
R. Králik et al., 2024	Sentinel lymph nodes in lateral compartment	Selective	Presence of metastasis
Luis Miguel Arciniegas et al., 2023	Sentinel lymph nodes (2-3 per case)	Routine	Presence of metastatic cells
Christopher R. McHenry et al., 1996	Thyroid nodule	Routine	1-6 sections per patient
H. Moon et al., 2009	Thyroid nodule	Routine intraoperative	Classified as malignant, indeterminate, defer, or benign

A critical distinction emerges between studies examining thyroid nodules versus lymph nodes. Studies focusing on lymph node frozen section predominantly targeted papillary thyroid carcinoma patients to assess nodal status and guide extent of neck dissection . In contrast, studies examining thyroid tissue primarily aimed to establish the diagnosis of malignancy to guide extent of thyroidectomy .

Decision Algorithms for Surgical Management

The studies employed various decision frameworks for modifying surgical plans based on frozen section results.

Extent of Thyroidectomy

Several studies described specific algorithms for determining whether to proceed from lobectomy to total thyroidectomy. Patients initially scheduled for thyroid lobectomy underwent completion thyroidectomy when frozen section identified metastases, malignancy, or specific high-risk features. Jiang et al. employed criteria including more than 5 lymph node metastases, or 1-5 metastases with extrathyroidal extension and/or multifocal carcinoma to prompt conversion to total thyroidectomy.

Extent of Lymph Node Dissection

For lymph node dissection decisions, multiple studies used frozen section of ipsilateral central neck nodes to determine whether bilateral central neck dissection was warranted. Chae et al. hypothesized that routine ipsilateral frozen section after elective ipsilateral central lymph node dissection, followed by contralateral dissection if positive, would reduce morbidity compared to routine bilateral dissection. Hartl et al. used frozen section of levels III and IV to determine whether to extend prophylactic lateral neck dissection to level II.

Decision Thresholds

Specific decision thresholds varied across studies. Min Jhi Kim et al. used a cut-off of 5 mm for nodal metastasis. Raffaelli et al. (2020a) employed a 2 mm threshold for distinguishing clinically significant metastases. Li Peng et al. established cut-off values of 2.050 mm for maximum metastatic diameter, 5.5 for total number, and 0.5342 for central lymph node metastasis ratio.

Diagnostic Accuracy of Frozen Section

The diagnostic performance of frozen section analysis varied substantially depending on the tissue examined and pathology type.

Frozen Section of Lymph Nodes

Study	Sensitivity	Specificity	PPV	NPV	Accuracy
M. Raffaelli et al., 2013	80.7%	100%	Not reported	Not reported	90%
Min Jhi Kim et al., 2021	94.6%	100%	100%	96.1%	Not reported
Yelda Jozaghi et al., 2013	68.8%	100%	100%	94.4%	Not reported
B. Chae et al., 2011	78.3%	100%	Not reported	Not reported	Not reported
M. Raffaelli et al., 2015	Not reported	Not reported	Not reported	Not reported	86%

Study	Sensitivity	Specificity	PPV	NPV	Accuracy
M. Raffaelli et al., 2020a	80.7%	100%	Not reported	Not reported	90%
D. Hartl et al., 2014	71.0%	99.6%	97.8%	92.4%	93.2%
M. J. Kim et al., 2018	93.5%	100%	100%	96.3%	97.5%

Frozen Section of Thyroid Tissue

Study	Sensitivity	Specificity	PPV	NPV	Accuracy
C. Bollig et al., 2018	95%	100%	100%	83%	Not reported
Ryan W. Berg et al., 2016	22%	Not reported	40%	Not reported	Not reported
R. Udelsman et al., 2001	3.4%	Not reported	Not reported	Not reported	Not reported
Y. Park et al., 2014	66%	99%	98%	87%	Not reported
Chaoyu Jiang et al., 2023	77.61%	100%	100%	88.46%	Not reported
S. Abu-Ghanem et al., 2016	65.7%	100%	100%	50%	74.5%
S. Olson et al., 2006	25%	Not reported	Not reported	Not reported	Not reported
Lei Zhang et al., 2007	95.9%	100%	Not reported	Not reported	98.4%
R. Callcut et al., 2006	67%	100%	100%	Not reported	96%
F. Lumachi et al., 2004	83.0%	100%	Not reported	Not reported	97.0%
L. Saydam et al., 2003	100%	87%	Not reported	Not reported	91%
Christopher R. McHenry et al., 1996	93%	100%	Not reported	Not reported	97%
A. Bastagli et al., 2001	42.9%	100%	100%	8.5%	92.0%
O. Prajapati et al., 2015	66%	99%	98%	87%	Not reported

Frozen section analysis of lymph nodes demonstrated consistently high specificity (99.6-100%) across studies, indicating that positive results reliably indicate true metastatic disease. Sensitivity ranged from 68.8% to 94.6%, with the

primary limitation being detection of micrometastases. Raffaelli et al. (2013) noted that five of five false-negative results were micrometastases (≤ 2 mm) , and Raffaelli et al. (2015) similarly found that five of seven undetected metastases were micrometastases .

For thyroid tissue analysis, accuracy was highly dependent on pathology type. Sensitivity was notably poor for follicular lesions, with Udelsman et al. reporting only 3.4% definitive diagnosis rate , Callcut et al. reporting 67% sensitivity , and Bastagli et al. reporting 42.9% . In 62-64% of follicular cases, the diagnosis was deferred to permanent histology . Conversely, frozen section performed well for papillary thyroid carcinoma diagnosis, with sensitivities ranging from 65.7% to 97.6% .

Impact on Surgical Decision-Making

Surgical Plan Modifications

Study	Modification Rate	Type of Modification	Appropriate Changes
Reema Mallick et al., 2019	5% (11/236)	Not specified	2.1% (5/236)
M. Raffaelli et al., 2020	26.6% (25/94)	Lobectomy to TT + bilateral CND	25 appropriate
R. Králik et al., 2022	46.6% (34/73)	Limited to more extensive surgery	Reduced two-step surgery risk
Ryan W. Berg et al., 2016	0.5% (1/192)	Lobectomy to completion TT	Not specified
R. Udelsman et al., 2001	3.4% (1/29)	Lobectomy to TT	Not specified
S. Olson et al., 2006	0% (0/135)	None	Not applicable
M. Raffaelli et al., 2015	65% (13/20)	Unilateral to bilateral CND	13 appropriate
M. Raffaelli et al., 2020a	16.6% (5/30)	Lobectomy to TT + bilateral CND	All appropriate
D. Hartl et al., 2014	59% (27/46)	Extension to level II	44% had metastases
A. Laurino et al., 2024	51.8% (57/110)	Lobectomy to TT	Not specified
Dena Arumugam et al., 2016	15.2% (12/79)	Lobectomy to TT	7/12 confirmed malignant
Christopher R. McHenry et al., 1996	3% (2/76)	Not specified	Not specified
A. Bastagli et al., 2001	3.9% (6/155)	Less extensive to TT	6 appropriate

The impact of frozen section on surgical decision-making varied dramatically across studies, ranging from 0% to 65% modification rates. Studies focused on lymph node assessment in papillary thyroid carcinoma consistently demonstrated higher rates of surgical modification. Raffaelli et al. (2020) found that frozen section identified occult nodal metastases in approximately 26.6% of patients scheduled for thyroid lobectomy . Similarly, Raffaelli et al. (2015) reported that 65% of pN1a patients in the ipsilateral central compartment dissection group had their surgical plan modified based on frozen section results .

In contrast, studies examining thyroid tissue frozen section for indeterminate nodules showed minimal impact. Olson et al. reported that in no patient over a decade did frozen section correctly alter intraoperative management

for multinodular goiter . Mallick et al. found that only 2.1% of frozen sections accurately changed the extent of thyroidectomy .

Prevention of Two-Stage Surgery

A key benefit identified across studies was the potential to avoid completion thyroidectomy as a second operation. Roach et al. calculated that without frozen section examination, 30 patients with indeterminate cytology would have required a second operation . Arumugam et al. found that 7 of 19 patients (37%) with malignancy avoided reoperation due to frozen section . Králik et al. (2022) demonstrated that frozen section of central lymph nodes reduced the need for two-step surgery compared to no frozen section (OR 1.93, p=0.026) .

Reduction in Unnecessary Surgery

Liu et al. demonstrated that a deep learning model applied to frozen sections reduced the rate of unnecessary lymph node dissection from 56.4% to 14.9% in clinically node-negative patients . Abu-Ghanem et al. reported that 12 patients did not undergo completion thyroidectomy due to benign final histology, appropriately avoiding unnecessary surgery .

Clinical Outcomes

Complications

Study	Transient Hypocalcemia	Voice Changes	Other Complications
R. Udelsman et al., 2001	Not reported	1 transient vocal cord dysfunction	None
B. Chae et al., 2011	Unilateral CLND: 31.8%; Bilateral CLND: 45.7%	Unilateral: 7.6%; Bilateral: 4.3%	No significant differences
M. Raffaelli et al., 2015	IpsiCCD: 18/50; BilCCD: 27/50	Not reported	No recurrent disease
M. Raffaelli et al., 2020a	6/13 (46%)	1 transient laryngeal nerve injury	No permanent complications

Several studies suggested that frozen section-guided surgical decision-making may reduce complications by avoiding unnecessary extensive surgery. Chae et al. found lower rates of transient hypocalcemia in the unilateral dissection group (31.8%) compared to bilateral dissection (45.7%) . Raffaelli et al. (2015) similarly reported more transient hypocalcemia in the bilateral group (27/50) versus ipsilateral plus frozen section group (18/50) .

Long-Term Oncologic Outcomes

Long-term oncologic outcomes were reported in limited studies. Raffaelli et al. (2015) reported no recurrent disease in either group . Raffaelli et al. (2020a) found no local or nodal recurrence at mean follow-up of 22.2 months . Hartl et al. found no cases of recurrent or persistent disease in level II even when extension of prophylactic lateral neck dissection to level II was not performed . Arciniegas et al. reported the percentage of reoperations for persistence and recurrence was significantly lowest in the sentinel lymph node group (2.04%) compared to controls (6.94% and 45.45%) .

Malignancy Detection Rates

Final pathology revealed malignancy in variable proportions across studies: 61.5% in suspicious cytology patients , 74.5% in Bethesda V patients , 17.5% in solitary nodule patients , 25% in large nodule patients , 34% in follicular neoplasm patients , and 13% in follicular neoplasm on frozen section .

Cost Analysis

Only a subset of studies addressed economic considerations.

Study	Cost Finding	Conclusion
C. Bollig et al., 2018	Savings of \$474 per case with routine FS	FS cost-effective for suspicious cytology
J. Roach et al., 2002	Savings of \$1,298 per patient	FS cost-effective for indeterminate cytology
R. Udelsman et al., 2001	\$12,470 per informative FS	FS not cost-effective for follicular neoplasms
R. Callcut et al., 2006	\$7,800 per useful FS vs. \$6,000 for completion thyroidectomy	FS not cost-effective for follicular lesions
Christopher R. McHenry et al., 1996	\$246-606 per patient, total \$26,040	Routine use not cost-effective
C. Bollig et al., 2018a	Savings of \$486 per case; \$768 excluding follicular lesions	FS cost-effective for large nodules

Cost-effectiveness of frozen section appears highly dependent on the clinical context. For patients with cytology suspicious for malignancy, routine frozen section resulted in cost savings of \$474-1,298 per patient . However, for follicular lesions where frozen section frequently defers diagnosis, the cost per useful frozen section ranged from \$7,800 to \$12,470, exceeding the cost of completion thyroidectomy . Berg et al. found that thyroid lobectomy without frozen section but with expedited overnight pathology was more cost-effective in their setting .

Synthesis

Explaining Heterogeneity in Frozen Section Utility

The dramatic variation in frozen section utility across studies (0-65% surgical plan modification rates) can be largely explained by three key contextual factors:

Pathology Type : Frozen section performs fundamentally differently based on the underlying pathology. For papillary thyroid carcinoma, where diagnosis depends on nuclear features, frozen section achieves sensitivities of 65.7-95% . For follicular neoplasms, where the diagnosis of carcinoma requires demonstration of capsular or vascular invasion, frozen section sensitivity drops to 3.4-67% , with 50-64% of cases deferred to permanent section . This biological constraint—the focal distribution of diagnostic features in follicular lesions—fundamentally limits frozen section utility in this context .

Tissue Examined : Studies examining lymph nodes in papillary thyroid carcinoma patients consistently demonstrated high utility with modification rates of 16.6-65% . Studies examining thyroid tissue for primary diagnosis

showed modification rates of only 0-3.4% . This difference reflects the distinct clinical questions being addressed: lymph node frozen section informs extent of neck dissection where preoperative imaging has limited sensitivity for occult metastases, while thyroid tissue frozen section attempts to establish diagnosis that may be more reliably determined preoperatively.

Preoperative Diagnostic Certainty : The utility of frozen section is inversely related to preoperative diagnostic certainty. When fine-needle aspiration is diagnostic for papillary carcinoma, frozen section did not alter management . When cytology is suspicious (Bethesda V), frozen section led to optimal operative procedure in 96% of cases . When cytology is indeterminate (Bethesda III-IV), frozen section provided useful information in only 47% of cases with indeterminate results .

Evidence-Based Recommendations by Clinical Context

High Utility Settings (frozen section recommended):

- Lymph node assessment in cN0 papillary thyroid carcinoma: Sensitivities of 68.8-94.6% with 100% specificity enable accurate staging and appropriate extension of neck dissection . This approach reduces morbidity compared to routine bilateral dissection while maintaining oncologic adequacy .
- Cytology suspicious for malignancy (Bethesda V): Frozen section leads to appropriate surgical extent in 96% of cases and achieves cost savings of \$474 per case .
- Large nodules (>4 cm) with non-follicular cytology: 50% of malignancies identified intraoperatively with cost savings of \$768 per case when follicular lesions are excluded .

Low Utility Settings (frozen section not recommended):

- Follicular neoplasms: The 3.4-67% sensitivity , 50-64% deferral rate , and cost of \$7,800-12,470 per useful frozen section make routine use neither informative nor cost-effective .
- Multinodular goiter: In one decade of experience, frozen section did not correctly alter management in any patient .
- Cytology diagnostic for papillary carcinoma: FNA was 97% accurate, and frozen section did not alter management .

Intermediate Utility Settings (selective use may be appropriate):

- Indeterminate cytology with clinical suspicion: Frozen section correctly identified malignancy in 47% of cases with indeterminate cytology, potentially avoiding second operations .
- Evaluation of extrathyroidal extension: Sensitivity of 66% with 99% specificity provides useful information for surgical planning in papillary thyroid carcinoma.

Micrometastases: A Consistent Limitation

Across all lymph node frozen section studies, micrometastases (≤ 2 mm) represent a consistent source of false-negative results . Raffaelli et al. (2013) found that 3 of 5 false negatives were micrometastases . Raffaelli et al. (2015) reported 5 of 7 undetected metastases were micrometastases . However, multiple authors note that micrometastases typically have limited clinical implications , and long-term follow-up studies show no recurrence even when micrometastases are missed .

Alternative Approaches

Several studies compared frozen section to alternative strategies. Expedited (overnight) pathology was proposed as an alternative to frozen section, potentially improving cost-effectiveness and patient quality of life. Liu et al. demonstrated that deep learning models applied to frozen sections outperformed ultrasound and CT for predicting lymph node metastasis (AUC 0.76-0.81 vs. lower values for imaging), suggesting technology may enhance frozen section utility. Molecular testing of thyroid nodules has reduced the role of frozen section, though this was not systematically evaluated in the included studies.

References

- A. Bastagli, L. de Pasquale, and L. Schubert. "[Follicular Neoplasm of the Thyroid: Usefulness of Extemporaneous Histologic Study of Frozen Sections]." *Annali Italiani Di Chirurgia*, 2001.
- A. Laurino, F. Pennestri, P. Procopio, A. Martullo, Gloria Santoro, P. Gallucci, F. Prioli, et al. "Impact of Nodal Status Evaluation on Therapeutic Strategy for Clinically Unifocal T1b/Small T2 Node Negative Papillary Thyroid Carcinoma." *Endocrine*, 2024.
- A. Shaha, and B. Jaffe. "Completion Thyroidectomy: A Critical Appraisal." *Surgery*, 1992.
- B. Chae, C. Jung, D. Lim, B. Song, Jeong Soo Kim, S. Jung, and J. Bae. "Performing Contralateral Central Lymph Node Dissection in Papillary Thyroid Carcinoma: A Decision Approach." *Thyroid*, 2011.
- C. Bollig, David R Gilley, D. Lesko, J. Jorgensen, T. Galloway, R. Zitsch, and L. Dooley. "Economic Impact of Frozen Section for Thyroid Nodules with "Suspicious for Malignancy" Cytology." *Otolaryngology Head & Neck Surgery*, 2018.
- C. Bollig, J. Jorgensen, R. Zitsch, and L. Dooley. "Utility of Intraoperative Frozen Section in Large Thyroid Nodules." *Otolaryngology Head & Neck Surgery*, 2018.
- Chaoyu Jiang, Yongzhao Chen, Chen Zhang, Bo Jiang, C. Qu, Guanqun Dong, Yanxue Wang, et al. "Intraoperative Frozen Section for Determining the Extent of Surgery in Papillary Thyroid Carcinoma: Comprehensive Risk Factor Assessment." *Gland Surgery*, 2023.
- Christopher R. McHenry, Christopher D. Raeburn, Theodore Strickland, and Jerry J. Marty. "The Utility of Routine Frozen Section Examination for Intraoperative Diagnosis of Thyroid Cancer." *American Journal of Surgery*, 1996.
- Christopher Wong, and M. Wheeler. "Thyroid Nodules: Rational Management." *World Journal of Surgery*, 2000.
- D. Hartl, A. Al Ghuzlan, I. Borget, S. Leboulleux, H. Mirghani, and M. Schlumberger. "Prophylactic Level II Neck Dissection Guided by Frozen Section for Clinically Node-Negative Papillary Thyroid Carcinoma: Is It Useful?" *World Journal of Surgery*, 2014.
- David F. Schneider, Linda M. Cherney Stafford, Nicole A. Brys, C. Greenberg, C. Balentine, Dawn M. Elfenbein, and Susan C. Pitt. "GAUGING THE EXTENT OF THYROIDECTOMY FOR INDETERMINATE THYROID NODULES: AN ONCOLOGIC PERSPECTIVE." *Endocrine Practice*, 2017.
- Dena Arumugam, C. Macias, Renee L. Arlow, Sumana Narayanan, Shou-En Lu, M. Deen, Michael May, et al. "Intraoperative Frozen Section May Reduce the Need for Reoperative Thyroid Surgery in Patients with Follicular Neoplasm," 2016.
- F. Lumachi, S. Borsato, A. Tregnaghi, F. Marino, A. Poletti, M. Iacobone, and G. Favia. "Accuracy of Fine-Needle Aspiration Cytology and Frozen-Section Examination in Patients with Thyroid Cancer." *Biomedicine & Pharmacotherapy = Biomedecine & Pharmacotherapie*, 2004.
- H. Moon, J. Kwak, Eun-Kyung Kim, Min Jung Kim, C. Park, W. Chung, and E. Son. "The Combined Role of Ultrasound and Frozen Section in Surgical Management of Thyroid Nodules Read as Suspicious for Papillary Thyroid Carcinoma on Fine Needle Aspiration Biopsy: A Retrospective Study." *World Journal of Surgery*, 2009.
- J. Paramo, and T. Mesko. "Age, Tumor Size, and in-Office Ultrasonography Are Predictive Parameters of Malignancy

- in Follicular Neoplasms of the Thyroid." *Endocrine Practice*, 2008.
- J. Roach, K. Heller, S. Dubner, and L. A. Szytler. "The Value of Frozen Section Examinations in Determining the Extent of Thyroid Surgery in Patients with Indeterminate Fine-Needle Aspiration Cytology." *Archives of Otolaryngology - Head and Neck Surgery*, 2002.
- L. Saydam, A. Kizilay, and M. Bozkurt. "Tiroit Nodüllerinin Değerlendirilmesi: Nce Ine Aspirasyon Biyopsisinden Sonra Frozen Section İncelemesi Gerekli Mi? The Evaluation of Thyroid Nodules: Is Routine Use of Frozen-Section Examination Necessary Following Preoperative Fine-Needle Aspiration Biopsy?" 2003.
- Lei Zhang, Wenya Li, and Mei-lin Jin. "[The Value of Frozen Section Examination in Thyroid Surgery]." *Lin Chuang Er Bi Yan Hou Tou Jing Wai Ke Za Zhi = Journal of Clinical Otorhinolaryngology, Head, and Neck Surgery*, 2007.
- Li Peng, Xiaoya Zheng, Ying Xue, Chun Huang, Xinliang Su, and Shanshan Yu. "Central Lymph Nodes in Frozen Sections Can Effectively Guide Extended Lymph Node Resection for Papillary Thyroid Carcinoma." *Annals Medicus*, 2023.
- Luis Miguel Arciniegas, V. Rekeň, Š. Nemergut, Martin Sabol, Štefan Durdík, and R. Králik. "The Importance of Sentinel Lymph Node Diagnostic Biopsy from the Lateral Neck Compartment in the Surgical Treatment of Papillary Thyroid Cancer." *Bratislava Medical Journal*, 2023.
- M. Haymart, D. Y. Greenblatt, D. Elson, and Herb Chen. "The Role of Intraoperative Frozen Section If Suspicious for Papillary Thyroid Cancer." *Thyroid*, 2008.
- M. J. Kim, C. Kim, Je Kim, and Young Sam Park. "Efficiency of Intraoperative Frozen Section Analysis of Central Neck Lymph Node Dissection in Patients with Papillary Thyroid Carcinoma." *International Journal of Surgery: Oncology*, 2018.
- M. Neale, L. Delbridge, T. Reeve, and A. Poole. "The Value of Frozen Section Examination in Planning Surgery for Follicular Thyroid Neoplasms." *Australian and New Zealand Journal of Surgery*, 1993.
- M. Raffaelli, C. Crea, L. Sessa, Piero Giustacchini, R. Bellantone, and C. Lombardi. "Can Intraoperative Frozen Section Influence the Extension of Central Neck Dissection in cN0 Papillary Thyroid Carcinoma?" *Langenbeck's Archives of Surgery (Print)*, 2013.
- M. Raffaelli, C. Crea, L. Sessa, S. Tempera, G. Fadda, A. Pontecorvi, and R. Bellantone. "Modulating the Extension of Thyroidectomy in Patients with Papillary Thyroid Carcinoma Pre-Operatively Eligible for Lobectomy: Reliability of Ipsilateral Central Neck Dissection." *Endocrine*, 2020.
- M. Raffaelli, C. De Crea, L. Sessa, G. Fadda, Chiara Bellantone, and C. Lombardi. "Ipsilateral Central Neck Dissection Plus Frozen Section Examination Versus Prophylactic Bilateral Central Neck Dissection in cN0 Papillary Thyroid Carcinoma." *Annals of Surgical Oncology*, 2015.
- M. Raffaelli, L. Sessa, C. De Crea, G. Fadda, P. Princi, E. Rossi, E. Traini, et al. "Is It Possible to Intraoperatively Modulate the Extent of Thyroidectomy in Small Papillary Thyroid Carcinoma?" *Surgery*, 2020.
- Min Jhi Kim, H. Kim, C. Park, and Bup-Woo Kim. "Frozen Section Analysis of Central Lymph Nodes in Papillary Thyroid Cancer: The Significance in Determining the Extent of Surgery." *Gland Surgery*, 2021.
- O. Prajapati, A. Verma, and M. Sabaretnam. "Intraoperative Frozen Section for the Evaluation of Extrathyroidal Extension in Papillary Thyroid Cancer." *World Journal of Surgery*, 2015.
- R. Callcut, S. Selvaggi, E. Mack, O. Ozgul, T. Warner, and Herb Chen. "The Utility of Frozen Section Evaluation for Follicular Thyroid Lesions." *Annals of Surgical Oncology*, 2006.
- R. Králik, E. Takácsová, I. Waczulíková, Miguel Arciniegas, Martin Sabol, Štefan Durdík, and M. Grigerová. "The Importance of Sentinel Lymph Node Intraoperative Frozen Analysis for Indication of Lateral Neck Dissection in Patients with Medullary Thyroid Cancer." *Neoplasma (Bratislava)*, 2024.
- R. Králik, M. Grigerová, E. Takácsová, M. Arciniegas, M. Sabol, V. Rekeň, Š. Nemergut, et al. "The Role of Intraoperative Frozen Section of Central Neck Lymph Nodes in the Management of Patients Preoperatively Diagnosed with Low- Risk Differentiated Cancer." *Rozhledy v Chirurgii*, 2022.
- R. Udelsman, W. Westra, P. Donovan, T. Sohn, and J. Cameron. "Randomized Prospective Evaluation of Frozen-

- Section Analysis for Follicular Neoplasms of the Thyroid.” *Annals of Surgery*, 2001.
- Reema Mallick, T. Stevens, T. Winokur, A. Asban, Thomas Wang, Brenessa Lindeman, J. Porterfield, and Herbert Chen. “Is Frozen-Section Analysis During Thyroid Operation Useful in the Era of Molecular Testing?” *Journal of the American College of Surgeons*, 2019.
- Ryan W. Berg, T. Yen, Douglas B. Evans, Bryan C. Hunt, F. Quiroz, Stuart D. Wilson, and Tracy S. Wang. “Analysis of an Institutional Protocol for Thyroid Lobectomy: Utility of Routine Intraoperative Frozen Section and Expedited (Overnight) Pathology.” *Surgery*, 2016.
- S. Abu-Ghanem, O. Cohen, Tom Raz Yarkoni, D. Fliss, and M. Yehuda. “Intraoperative Frozen Section in “Suspicious for Papillary Thyroid Carcinoma” After Adoption of the Bethesda System.” *Otolaryngology Head & Neck Surgery*, 2016.
- S. Olson, Yusra Cheema, J. Harter, J. Starling, and Herb Chen. “Does Frozen Section Alter Surgical Management of Multinodular Thyroid Disease?” *Journal of Surgical Research*, 2006.
- Y. Park, Soo-Geun Wang, J. Goh, D. Shin, In-Ju Kim, and Byung-Joo Lee. “Intraoperative Frozen Section for the Evaluation of Extrathyroidal Extension in Papillary Thyroid Cancer.” *World Journal of Surgery*, 2014.
- Yelda Jozaghi, Keith Richardson, S. Anand, A. Mlynarek, M. Hier, V. Forest, E. Sela, M. Tamilia, D. Caglar, and R. Payne. “Frozen Section Analysis and Sentinel Lymph Node Biopsy in Well Differentiated Thyroid Cancer.” *Journal of Otolaryngology - Head & Neck Surgery*, 2013.
- Yihao Liu, Fenghua Lai, B. Lin, Yunquan Gu, Li-li Chen, Gang Chen, Han Xiao, et al. “Deep Learning to Predict Cervical Lymph Node Metastasis from Intraoperative Frozen Section of Tumour in Papillary Thyroid Carcinoma: A Multicentre Diagnostic Study.” *EClinicalMedicine*, 2023.